

CROSS CUTTING MANAGEMENT PLAN (NWFP)
for
MAMUNG PAKSHING TSHONGDREL TSHOGPA
(1st Revision)



Plan period: January 2025 to December 2035

Park Office: Royal Manas National Park, Gelephu

Gewog: Phangkhar

Dzongkhag: Zhemgang

Table of content

APPROVAL SHEET.....	III
PART A: NWFP MANAGEMENT AND MARKETING PLAN	1
Background	1
Goals and Objectives are:	1
Description of NWFPs present in the area demarcated for management.....	1
Harvesting prescriptions for sustainable NWFP harvesting.....	3
Participatory Mapping and Google Maps showing the different NWFP Harvesting Areas.....	4
Marketing and Sales Plan	5
Marketing Arrangements.....	6
Benefits to the Members.....	7
Benefits to the group.....	8
PART B: GROUP'S BY-LAW	9
NWFP management&MarketingGroup.....	9
Executive committee of NWFP management and marketing group	9
Tenure of Committee Member	10
Incentives to the committee members	10
Sources for generation of group fund	10
Group fund management and utilization.....	11
Meeting and Purpose	11
Penalties.....	11
Annexure 1: Membership Lists	12

APPROVAL SHEET

FRPMD/FRPIS_NWFP/01/2024/565 (30-12-2024)

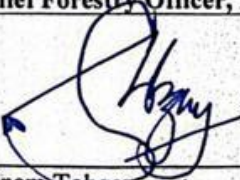
APPROVAL SHEET

Name of NWFP Management and Marketing Group	Mamung Pakshing Tshongdrel Tshogpa
Village(s)	Mamung
Gewog	Phangkhar
Dzongkhag/Dungkhag	Zhemgang
No. of households	16
Area (ha)	38.63 ha
Plan period	January 2025-December 2035

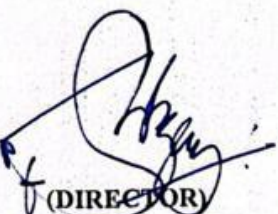
Prepared and submitted for approval:

	Chairperson of the NWFP Management and Marketing Group	Gup of the Gewog	Head FRPM section of Division/Park	Chief Forest Officer of Division/Park
Signature:				
Name:	Tshering Dhendup	Tshering Dorji	Tshering Dorji	Samten Wangchuk
Date:	18/09/2024	4/10/2024	10/10/2024	10/10/2024

Reviewed and recommended for approval

	Reviewing officer, FRPMD	Chief Forestry Officer, FRPMD
Signature		
Name	Kezang Wangmo	Sonam Tobgay
Date	11-12-2024	18/12/24

Approved by:


(DIRECTOR)

Department of Forests and Park Service

PART A: NWFP MANAGEMENT AND MARKETING PLAN

1. Background

The cross-cutting management plan is prepared and developed for sustainable management of non-wood forest products falling within the protected area-based plan of Royal Manas National Park. The cross-cutting management plan is developed for Mamung Pakshing Tshongdrel Tshogpa for Pantang-Mamung Chiwog of Phangkhar Gewog, Zhemgang Dzongkhag.

Mamung Pakshing Tshongdrel Tshogpa is located in the Pantang-Mamung *chiwog* of Phangkhar *gewog*, Zhemgang *Dzongkhag*, comprising 16 households as members. The community primarily generates income from the sale of cash crops such as mandarins, cardamom, and ginger. The area ranges from 340 to 860 meters above sea level and features sub-tropical forests, with bamboo (*Dendrocalamus hamiltonii*) being the dominant species. The boundary location of the NWFP areas are as follows: 1) Yaling at the western side: 26.988290, 90.817229, 2) Khashaling at the northern side: 26.995922, 90.814042, 3) Kirangang at the southern side; 26.989532, 90.839131, 4) Oling at the eastern side above highway road; 26.995058, 90.839131. The total areas identified as NWFP collection sites is 38.63ha.

On average, each household earns annual income from agriculture products, about Nu 50,000 to Nu 300,000 Ngultrum per year. Currently, the Mamung NWFP group has primarily focused on selling bamboo culms for construction and bamboo shoots in nearby markets. Members have gained significant knowledge in the sustainable harvesting of bamboo and have also learned some processing techniques for bamboo shoot pickle making for sale.

Review of the previous plan and the way forward:

This marks the third proposal for a plan revision, aimed at achieving the objectives set in previous plans, which were not fully realized due to challenges such as the COVID-19 pandemic and marketing issues. In their initial phase, the group successfully marketed bamboo shoots and culms to local buyers, generating an income of around 53,000 Ngultrum.

The motivation behind renewing the group includes the desire to enhance income through collective action and to address marketing challenges. The members foresee opportunities to expand their market reach to upcoming projects, such as the Chamkharchu Project and the Gelephu Mindfulness City development, as well as other NWFP dealers.

The revision of the plan is felt crucial to achieve the objectives of the earlier plans while also exploring additional NWFPs and new areas for sustainable management of resources. The group aims to leverage their experiences and knowledge gained to enhance their marketing strategies and increase their overall income.

2. Goals and Objectives are:

- To generate income for the people of the Mamung community through the sale of bamboo shoots, bamboo culms and sale of other NWFPs.
- To sustainable harvesting and marketing of Bamboo resources along with other NWFPs.
- To create employment opportunities within the communities.

3. Description of NWFPs present in the area demarcated for management

There are around 10 collection areas specified for the collection of listed NWFPs based on the resources availability in those identifies areas. Various NWFPs grows during the specific season

as mentioned below in Table 1. There are about 13 major products listed as per the commercial values

Table 1: Brief description of NWFPs for commercial harvesting in each season.

NWFP name	Harvesting area(s) (name)	Parts harvested	Use/ purpose (traditional & others)	Harvesting season (months)	Estimated quantity for harvesting season/year
Bamboo Culm (Paibing)	Paplang, Pangleng, Yurtibi, kirangang 1&2, Angala, tekari, oiling,	Culm	Fencing, Handicrafts, Constructional purposes, etc	Dec-Jan	25000 culms
Bamboo shoot (Bungba)	Paplang, Pangleng, Yurtibi, kirangang 1&2, Angala, tekari, oiling,	Shoot	Vegetable, Pickle	Jul-Aug	20000 kg
Pipla (<i>Pipla</i>)	Paplang, Pangleng, Yurtibi, Kirangang 1&2, Angala, Tekari, Oiling, Kerchlaa, Kashaling	Fruit	Medicinal values	Oct-Nov	1000 kg
Fern (Nakey)	Kerchla & Bayling, Dhumtii, Manling gang	Young shoot	Vegetables	Feb-Mar-Jul	1500 bundle
Dhamru (Dhamru)	Bayling & Kerchala	Leaves	Vegetable	Apr and Oct	1000 bundle
Wild Potato (surdhangKii)	Khashaling & Renjang	Tuber	Food	Dec & May	1000 kg
Sapindusmukro ni (nakapani)	Kerchala, Khashaling	Fruit	Medicine	Oct-Nov	1000 kg
Litsea cubeba (Pharchakpa)	Mamung locality	Fruit	Medicine	Jul-Aug	1000 kg
Cane shoot (Patsa)	Manling	Shoot	Vegetable	Apr-Jun	500 pcs
Banana flower (muchali)	Manling, Gothpabrang	Flower	Vegetable	Feb-Mar	2000 pcs
Cannarium resin (poikar)	Kerchala	Resin	Incense	Jul-Aug	100 kg
Mushroom (mu)	Yaling, Kerchala, Manling	Fungus	Vegetable	Feb-Jul	500 kg

Table 2: Available bamboo resources, as per the assessment result.

Block areas	Year/age category		
	Year-1 (culms)	Year-2 (culms)	3 years above (culms)
Kerigang	1919	1212	8585
Oyeleng	2208	1536	6192
Tekarey	3450	1550	10450
Yurtibi	1388	2066	5044
Paplang	4500	4100	11900
Pangleng	1900	1300	7050
Angala	900	1800	4600
Total (nos/culms)	16265	13564	53821

The bamboo assessment was done based on old CF manual2 (plot size was 25mx10m, and the plots were randomly selected)

4. Harvesting prescriptions for sustainable NWFP harvesting

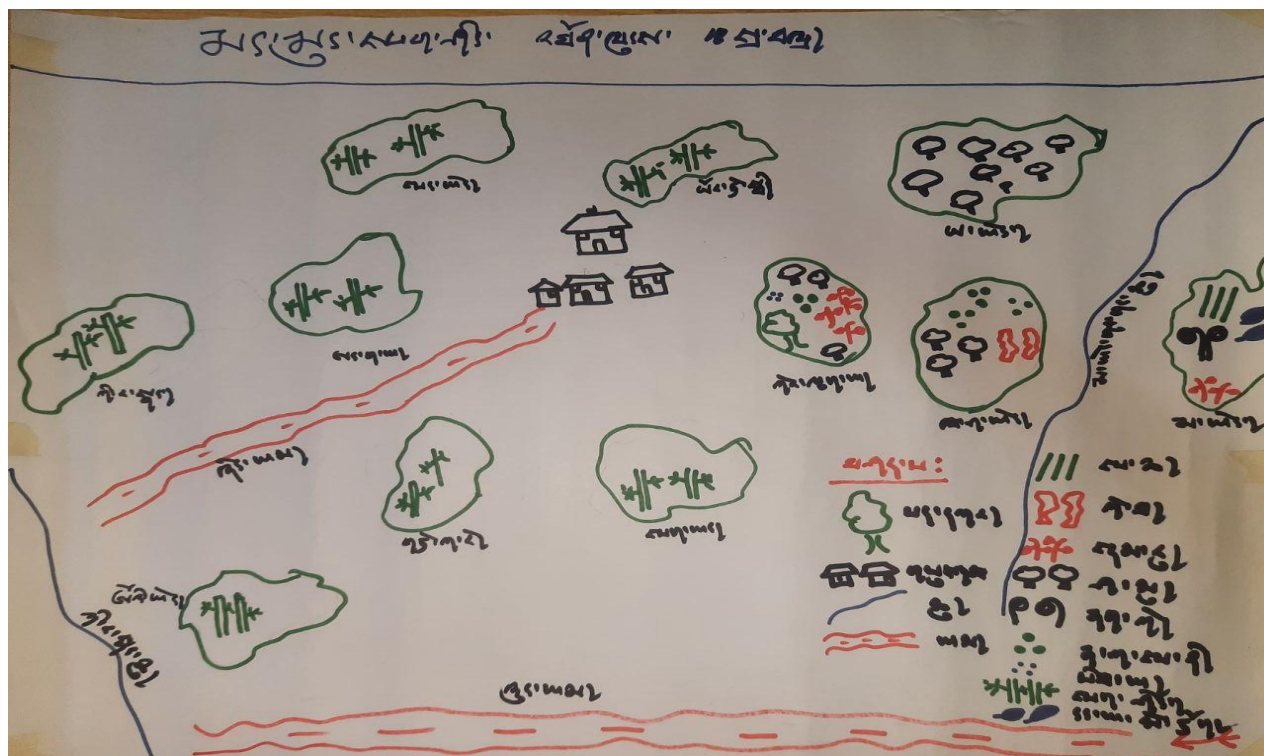
To make the harvesting of NWFP sustainable, the MAMUNG PAKSHING TSHONGDREL TSHOGPA has to apply an appropriate harvesting prescription based on scientific and local knowledge. The chairman will work according to the seasonal calendar and the period of the guidelines. During the harvesting season, the chairman will collect a permit from the Gomphu Range office for the required head loads in the name of the NWFP group and then inform the members to proceed with harvesting the resources. The harvesting areas will be identified to the members as per the above table No. 1. The monitoring works will be also done by the Committee members for the proper harvesting as per the guidelines.

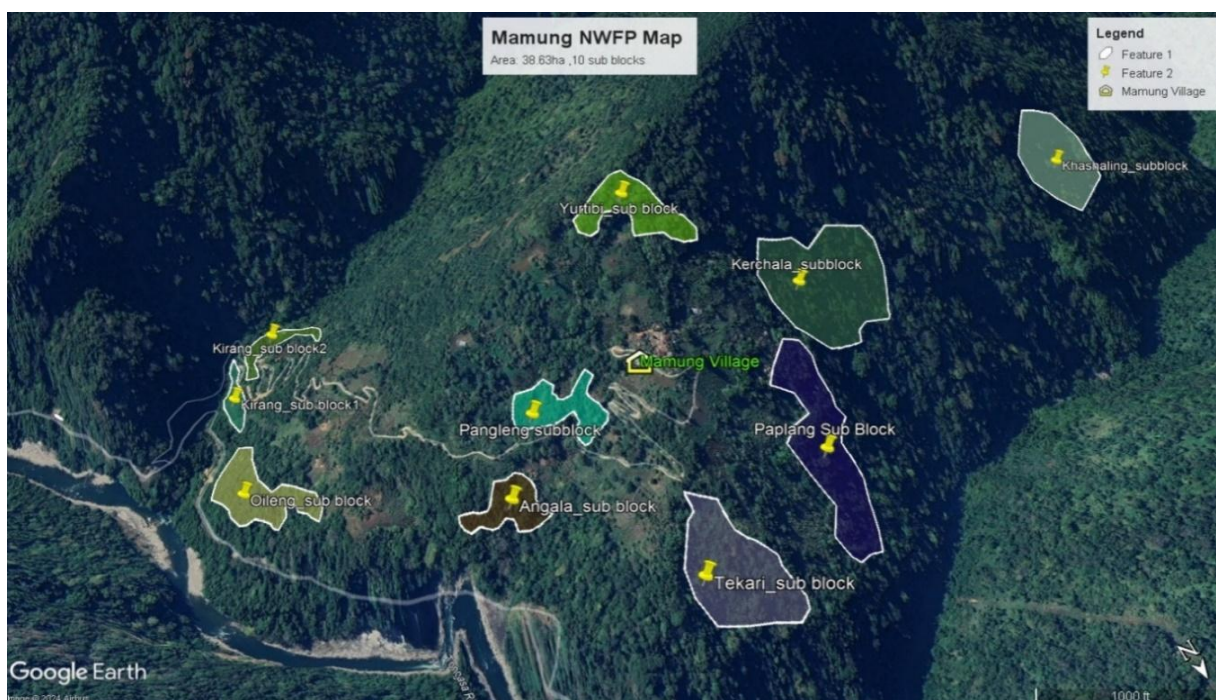
Table 3: Prescriptions for sustainable harvesting of NWFPs

NWFP species	Harvesting prescriptions for sustainable harvesting
Bamboo culm and bamboo shoot (<i>Dendrocalamus hamiltonii</i>)	✓ Cut culms above one node from the ground level ✓ Do not cut culms less than one year old ✓ Retain 50% of healthy culms in a clump ✓ Remove matured culms from the centre of the clump. ✓ Do not harvest the culms growing on the periphery of the clump ✓ Collect bamboo shoots to a maximum of 25% from a clump ✓ Cut bamboo shoots with a sharp knife at the ground level without damaging or wounding the other healthy ones ✓ After cutting and collecting the bamboo shoot, cover that void place with topsoil Protect the remaining shoots and let them further develop into culms
Pipla (<i>Piper pedicelatum</i>)	✓ Do not uproot or cut down the entire climber. ✓ Collect fruits only when fully mature. ✓ Harvest fruits only after the announced starting date for harvesting (The date is usually announced by the coordinator / relevant authority). Harvest fruits/spikes ✓ by hand picking without damaging the plant or by cutting a few fruit-bearing branches. ✓ Do not cut down the host tree to ease harvesting. ✓ Do not cut the entire fruiting portion of the plant
Nagkey (<i>Diplazium esculentum</i>)	✓ Harvests young shoots that are 15cm in length. ✓ Don't harvest all shoots coming up in season. ✓ Retain at least 50% of the young shoot of the last flush for the development of leaves. ✓ Don't harvest the shoots that have developed into leaves.
Dhamru (<i>Elatosteme lineolatum</i>)	✓ Collect plants that are 15cm and above. ✓ Retain stump of around 3" for sprouting of new shoots. ✓ Don't uproot whole plant. ✓ Shouldn't cut down trees and create more openings. ✓ Shouldn't disturb its catchment area.
Surdhangkii (<i>Dioscorea spp</i>)	✓ Manual digging hole with the help of a hoe. ✓ select the mature climbers ✓ keep the upper part of the tuber for regeneration.

Nakapani (<i>Sapindus mukorossi</i>)	✓ Collect nuts by shaking the tree and handpick from the forest floors ✓ If nuts are still on the tree use long bamboo poles to detach it. ✓ Do not fall trees to ease harvesting and collect about 75% of the fruits and retain 25 % for ecological circle.
Pharchakpa (<i>Litsea cubeba</i>)	✓ Harvest fruit by handpicking or by lightly lopping off outer branches. ✓ Do not cut down the tree for harvesting of fruit.
Patsha (<i>Plectocomia himalayana</i>)	For cane shoots: ✓ Cut young shoot by bending the cane stem with the help of a hook long and strong stick using sharp knife. ✓ Do not cut tree climbed cane for shoot. ✓ Care shall be taken to avoid damaging sucker buds.
Poikar (<i>Canarium strictum</i>)	✓ Collect the solidified resin with the help of a knife. ✓ Do not damage all the taping areas while collecting the product. ✓ blazing only 50% of the bark periphery.
Muchali (<i>Musa sp</i>) /Banana flower.	✓ With the help of a long stick try to pluck the flower ✓ Do not collect all the flowers from came location ✓ Never cut the whole banana tree

5. Participatory Mapping and Google map showing the different NWFP Harvesting Areas





6. Marketing and Sales plan

In the previous plan periods, the NWFP group members of Mamung couldn't market products as planned, so they could generate very little income. The main reason behind this is that during that term (2019-2022), the COVID-19 pandemic has hampered their sales plan, both for locals and outsiders. For the present plan, the following products listed will be the main products that will be sold in the market based on the demand made by the customers. The sale plan was formulated based on the interim framework for management and marketing of NWFP guidelines 2022. The prices are fixed based on the prevailing local market rates as tabulated under.

Table 4: Marketing and sales plan of the group

NWFP	Potential market/buyer	Estimated quantity for sale per season (unit)	Form of Products (dry/fresh)	Expected price per unit	Total (Nu)
Bamboo culm	Chamkharchu project Local Contractors Upcoming GMC project	25000culm	culm	65	1625000/-
Bamboo shoot	KNC, GMC project Gelephu	3500kg	Fresh shoot	35	87500/-
Pipla	NWFP Dealer	1000kg	Dry fruit	100	100000/-
Fern	Local Market	500bundle	fresh shoot	20	10000/-
Dhamru	Local Market	500bundle	Fresh shoot	30	15000/-
Wild potato	Local Market	1000kg	Fresh tuber	150	150000/-
Nakapani	NWFP Dealer	1000kg	Dry fruit	80	80000/-
Pharchakpa	NWFP Dealer	500kg	Dried fruit	125	62500/-

Patsa	Local market	200 pcs	shoot	80	16000/-
Poikar	NWFP Dealer&LocalMarket	100kg	Solidified resin	300	30000/-
Muchali	Local Market	500pcs	Fresh Flower	50	25000/-
Mushroom	Local Market	500kg	Fresh	150	75000/-
				Total amount	2,276,000/-

7. Marketing Arrangements

NWFP Products such as Bamboo shoots and mushrooms in smaller quantities can be sold in Pantang and Panbang markets. For NWFP with larger quantities, can be aimed to be sold in the Gelephu market and other project areas. In order to confirm the smooth functioning of the business, the dealer from the market will be arranged by the committee members of the group.

The Chairman will monitor and explore the demand from nearby institutes, agents, companies and other business groups. For further arrangements, support will be sought from the DAMC. In addition, the group will focus more on diversified handicraft development and will explore for other innovative ideas which have good scopes in the market.

Activity	A brief description of how the activities are carried out
Harvesting of products	The members will harvest the NWFPs in groups, market it together in teams and share the income from the sale of the product according to the by-laws framed in Part 2.
Post-harvest practices	The members will try to prepare the products for sale, and all the grading works and packaging etc will be done in groups as per the Chairperson's instructions.
Marketing of products	The NWFP collected and harvested will be marketed in groups as per the by-laws.
Identification of buyers	The chairperson will link with the licensed buyers
Price fixing	The price fixing will be as per the marketing and sales plan. The Chairperson will fix the price according to the current market value.
Income sharing	50% of the income will be deposited in the group account and the remaining 50 % will be shared among the members after realizing all the expenses incurred during the work.
Raising of community development fund	50% of income from the sale of NWFP, Fee collection, and other collections will be deposited in the BDBL group account CDF.
Payment to committee members for their service	Committee members will be given a vehicle charge and daily allowance of Nu500 per day while visiting other places for the NWFP-related works from the group fund and during the normal group works equal share will be given to them like members.

8. Benefits to the Members

The benefit or income from the sale of Non-Wood Forest Products will be equally shared among the active members who have worked for the collection, transportation, seasoning and marketing of the products. The financial projections were done for only potential NWFPs.

Table. 6 Financial projections of benefits to the members from NWFP sales

Sl. No	Particulars	Products(names)				Total Amount (Nu)	remarks
		B.culms	B.shoot	pipla	Shamu		
1	Revenues						
a	Total quantity(unit)	25000nos	3500kg	1000kg	500kg		
b	Market price(Nuperunit)	65/-	35/-	100/-	150/-	-	
c	Totalrevenuesfromforest product sales (a*b)	1,625,000/-	87500/-	100000/-	75000/-	1,887,000/-	
A	Total revenues	1,625,000/-	87500/-	100000/-	75000/-	1,887,000/-	
2	Expenses						
b	Royalty	50000/-	7000/-	20000/-	2000/-	79000/-	Royalty on Forests Produce 2006
e	Contribution to community development fund	812,500	43750	50000	37500	943,750	
B	Total expenses	862500	50750	70000	39500	943,750	
C	Total benefit(net income) of group members (Total revenue–Total expenses B)	762500	36750	30000	35500	943250/-	
D	Average income to group members(C/n ofgroup members)	47656	2296	1875	2218	54045	
E	Price per unit paid to group members (C/1a)	1.90	0.65	1.87	4.43	8.85	

*** Royalty rates are based on the FNCR 2006.

9. Benefits to the group

The 50% of the income generated from the sale of NWFPs will be shared among the members after the sale of the product upon calculating all the expenses incurred for the work and 50% will be deposited in the group's account as a main source for the group fund development. Below in the table is only 50% of the income from the sale of NWFPs, collection of fees and fines that will be saved as the main fund(CDF) for the group. The projection below is done only for major/potential products that will generate more income for the Group.

Table: Financial projection of the benefits to the group, the community development fund

Sl. No	Particulars	Amount(Nu)
	Revenues	
1	Membership registration fees	8000/-
2	Annual contribution	1600/-
3	Contribution to the Community Development Fund (CDF)from sales of Bamboo culm	812500/-
4	Contribution to the CDF from sales of Bamboo shoot	43750/-
5	Contribution to the CDF from sales of Pipla	50000/-
6	Contribution from the CDF from sales of Shamu	37500/-
A	Total revenues	953,350/-
	Expenses (only include expenses that will be paid from the group funds)	
	Expenses for the opening of Joint account	5000/-
	Refreshment for harvesting of products(in lump sum per season)	50000/-
	Permit fees	79000/-
B	Total Expenses	134000/-
C	Balance: Total revenues–Total expenses	819,350/-

PART B: GROUP'S BY-LAW

10.NWFP management & Marketing Group	
10.1 Membership Arrangement and resignation 10.2. Responsibilities and rights of members:	<u>Membership arrangement:</u> • All the households residing in Mamung village are eligible to join as members of the management group; • Acceptance in the group will be made only by the committee members through the general assembly.
	<u>Procedure for becoming new members:</u> • Any individual wishing to become a member of the group will put up a letter/application to the chairperson; • A member wishing to join the group should be 18 years old and above, must be Head of the Household, he or she should be permanent residence of Mamung village; • A person wishing to join the group as new member, from the fragmented families of the villages /can become a member by paying Nu: 5000/- as a membership fee.
	<u>Procedure for leaving the group:</u> • If anyone from the group wishes to discontinue from a group member, he/she has the right to leave. However, he/she should leave with the submission of the application to the chairperson and will not be allowed to join the group in the future. Further, the member leaving the group will not be allowed for any kinds of benefit from the group fund. • He/she can leave the group after discussing with the committee members; about paying a fine of Nu 500/-
	<u>Responsibilities and rights of members:</u> • Implement activities approved by the government that have been planned in their management plan. • Attend ad-hoc meetings conveyed by committee members. • Protect illegal activities and refrain from doing illegal activities in the collection area. • Harvest the bamboo only as per the management plan.
11. Executive committee of NWFP management and marketing group	
11.1 Chairperson	<u>Responsibilities and rights of members:</u> Name: Tshering Dhendup Contact: 17631953 • Implement activities approved by the government that have been planned in their management plan. • Attend ad-hoc meetings conveyed by committee members. • Share the minutes of meeting and resolution to the group members • Monitor the illegal activities that may happen in the identified collection areas

	• Advise the group members to refrain from doing illegal activities in the collection area. • Harvest the bamboo only as per the management plan.
11.2 Secretary	<u>Responsibilities of Secretary:</u> Name of Secretary: Karma Wangchuk Contact: 17539057 • Obtain a collection permit for the product from the Range office. • Officiate as Chairperson in his absence. • Assist the chairperson in the planning of activities. • Maintain records of any activities conducted in the group. • Maintain a record of tools and implement office correspondence.
11.3 Treasurer	<u>Responsibilities of Treasurer:</u> Name of Treasurer: Dechen Lhamo and Rinchen Wangmo Contact: 17849374 • To maintain records of cash transactions such as: - Collection and deposition of fines in the group account - Collection and deposition of annual membership fee - Recording of expenditure for meeting and labour engagement - Recording of expenditures for group activities - Issuance and recording of receipts for fine collection - Share the details of fund accumulation to members during meetings
12. Tenure of committee member	<u>Terms for Committee members:</u> • All the committee members like Chairperson, Secretary and Treasurer will be for 5 years. They will be re-elected after five years if capable of moving forward
13. Incentives to the committee members	<u>Incentives to the committee members:</u> • Transportation charges, Food, and Refreshments as per the prevailing local rate will be provided during any activities related to the group from the group income/fund.
14. Sources for generation of group fund	

14.1 Membership registration fees 14.2 Annual contribution 14.3 Collection from sale proceeds	<u>Sources of funds:</u> • New membership fees nu 500 • Annual fees nu 100 from all the members • In order to raise the funds and meet other activities 50% of the sale of products will be deposited in the joint savings account (CDF).
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15. Group fund management and utilization	<u>Group Fund Utilization:</u> • NWFP resource development • Welfare to members during natural calamities. • Community cultural activities. • For refreshment purposes during the group activities. • The funds in the bank will be withdrawn twice by the committee members jointly (after 5 and 10 years respectively).
16. Conflict resolution	<u>Conflict Resolution:</u> • In case of disputes or conflicts among communities arise about access to certain areas for NWFP harvesting; the matter shall be put to the village Tshogpa, Gewog Tshogchung (GT) Dzongkhag Tshogdue (DT) for further decision. • Conflicts or disputes within the NWFP Management Group, which cannot be resolved by the group members shall be forwarded to the coordinator for settlement. If the case cannot be resolved, it shall be forwarded to the Gewog Administration. • In case there is a conflict between the NWFP Group and outsiders, the case shall be forwarded to the Gewog Administration by the Coordinator and Forest Range office.
17. Meeting and purpose	<u>Meetings:</u> To plan and discuss an important activity for the management of NWFP, meetings will be conducted as follows.
a. General meeting (all group members)	To be conducted every 2nd, 6th and 11th months of Bhutanese calendar (discuss the planned activities before the harvest of products)
b. Committee members meeting/ad-hoc meeting	To be conducted as and when required to discuss the activities planned and to discuss the progress of group activities.
18. Penalties	<u>Penalties:</u> • Absentees during working and meetings shall be fined Nu: 400/- each time; • Violation of the NWFP business plan shall be fined according to the quantity misused, as per the detailed inspection done by the committee members. • Collection of NWFPs during the off-season shall be fined double the market price, each time; • If any of the members sells the bamboo resources he/she will charged the market price scheduled above in Table 3.

Annexure 1: Membership Lists

Membership list of Mamung Non wood Forest Product Group

Sl.no	Name	CID no	Village	Gung no	Designation	Sign	Thru No
1.	Sonam Jambay	12006000 321	Mamung	Wa-6-95	Members		119
2.	Tshering Dendup	12006000 263	Mamung	Wa-6-81	chairman		266
3.	Yonten Tobgay	12006000369	-11-	Wa-6-88	Members		122
4.	Leki	12006000 329	-11-	Wa-6-86	Member		175
5.	Dorji Tshering	12006000 277	-11-	Wa-6-82	Member		213
6.	Nim Dorji	12006000 346	-11-	Wa-6-87	Member		121
7.	Cheban	12006000 439	-11-	Wa-6-94	member		144
8.	Nima Gyelpo	12006000 390	-11-	Wa-6-90	Accountant		Nil
9.	Jampel Yonten	12006000 321	-11-	Wa-6-85	Member		145
10.	Tshering Norbu	12006000 213	-11-	Wa-6-78	Member		118
11.	Gyempo Lhamo	12006000 432	-11-	Wa-6-93	Member		107
12.	chungzer	12006000 412	-11-	Wa-6-91	Secretary		48
13.	Lodhen	12006000 293	-11-	Wa-6-83	Member		179
14.	Dorji Wangchuck	12006000 384	-11-	Wa-6-89	Accountant		117
15.	Tshering chodhen	12006000 241	-11-	Wa-6-80	Member		114
16.	Dorji Tshering (Lichila)	12006000 420	-11-	Wa-6-92	Member		Nil
47.							